

Chem Balancer II

a program that balances chemical equations on your TRS-80 ModI LII & Mod III.

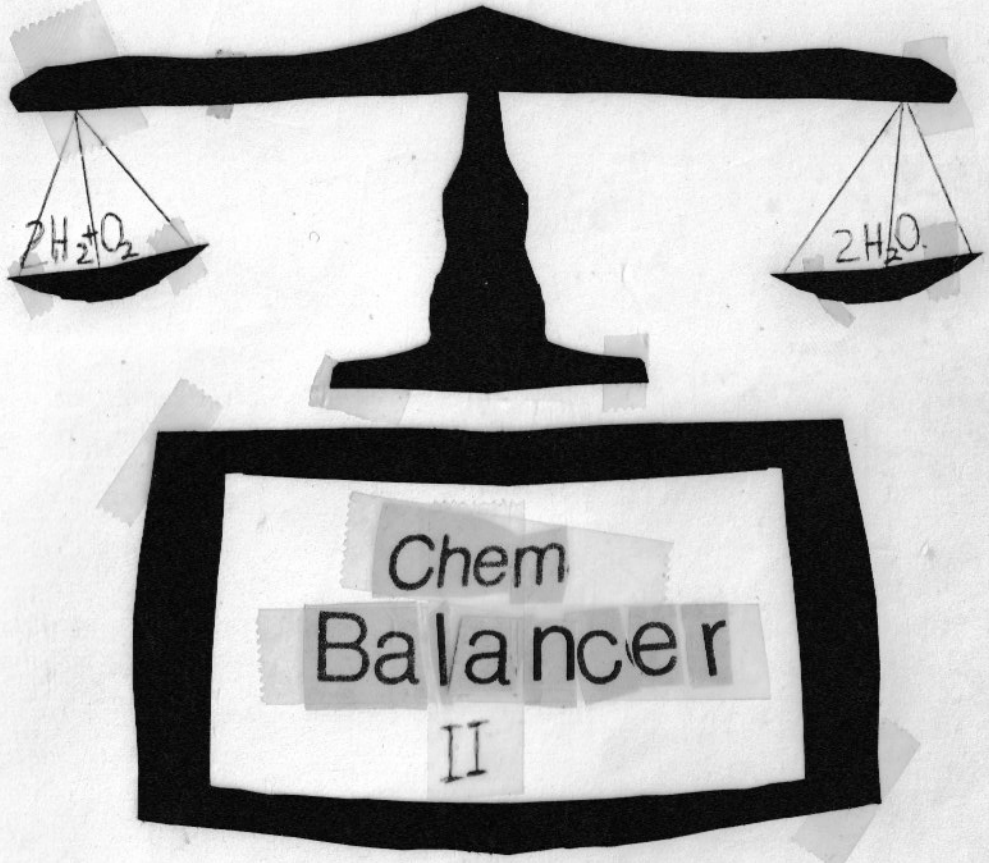
,Kadie Software

Chem Balaner II

Kadie Software

[redacted]

CHEM BALANCER II © 1981 Carl Kadie



A CHEMICAL EQUATION BALANCER for the TRS-80, Model I Lev II or Model III. Requires at least 16K

© 1981 Carl Kadie
*a trademark of the Tandy Corporation

Copyright Notice
© 1981 Carl Kadie, Peoria, Illinois
It is a violation of the law to reproduce these materials for any purpose other than personal convenience.

ALL KADIE SOFTWARE COMPUTER PROGRAMS ARE DISTRIBUTED ON AN "AS IS" BASIS WITHOUT WARRANTY

Kadie Software shall have no liability or responsibility to customer or any person or entity with respect to any liability, loss or damage caused or alleged to be caused directly or indirectly by computer programs sold by Kadie Software, including but not limited to any interruption of service, loss or buisness or anticipatory profits or consequential damages resulting from the use or operation of such computer programs.

<page 1 missing>

<page 2 missing>

* easy entry of raw equations.

* the ability to balance simple, double, triple, and quadruple replacement reactions.

* great accuracy and precision because fractions are stored as fractions with 32 digits of precision. All of which makes erroneous, overflowed, underflowed, and indeterminate answers virtually unheard of.

* and many more.

LOADING AND RUNNING INSTRUCTIONS

1. Put the ChemBalancer II tape in the tape player with with the tape's side one up.

2. Rewind the tape to its beginning.

3. Connect the black plug to the EAR socket of the tape player. Make sure that the two grey plugs are not connected to the tape player.

4. Type CLOAD and press ENTER.

5. Press the PLAY button on the tape player.

6. In a few moments two asterisks (**) will appear and

* easy entry of raw equations.

* the ability to balance simple, double, triple, and quadruple replacement reactions.

* great accuracy and precision because fractions are stored as fractions with 32 digits of precision. All of which makes erroneous, overflowed, underflowed, and indeterminate answers virtually unheard of.

* and many more.

(ex #3) Hypothetical triple replacement

1. $A_3E_3I_2 + D_{21}H_8C_{21} + G_8B_{42}F_{21} \rightarrow A_2B_3C + D_2E_4F_3 + G_3H_2I_7$
2. A3E3I2 + D21.H8.C21 + G8.B42.F21 = A2.B3.C + D2.E4.F3 + G3.H2.I7
3. 28(A3E3I2) + 2(D21.H8.C21) + 3(G8.B42.F21) → 42(A2.B3.C) + 21(D2.E4.F3) + 8(G3.H2.I7)

5¾ m

(ex #6) Cloride ion is oxidized to clorine gas

1. $MnO_4^- + H^+ + Cl^- \rightarrow Mn^{2+} + Cl_2 + H_2O$
2. MNO4@- + H@ + CL@- = MN@2 + CL2 + H2O
3. 2(MNO4@-) + 16(H@) + 1Ø(CL@-) → 2(MN@2) + 5(CL2) + 8(H2O)

3½ min

(note: MnO_4, could have, but doesn't need a period between the MN and the O4, because MN is two letters long.)

(ex #7) Oxidation of iron

1. $Cr_2O_7^{2-} + H^+ + Fe^{2+} \rightarrow Cr^{3+} + Fe^{3+} + HOH$
2. CR2O7@-2 + H@ + FE@2 = CR@3 + FE@3 + H.O.H
3. (CR2O7@-2) + 14(H@) + 6(FE@2) → 2(CR@3) + 6(FE@3) + 7(H.O.H)

4 5/6 min

the right one will blink repeatedly on and off, if the tape is loading correctely.

7a. Approximatly four minutes later

b. the screen will display:

READY

>_

Press the [stop] button on the tape player.

8a. Now type RUN and press [ENTER].
If the tape has loaded correctly

b. the screen will display:

CHEMICAL EQUATION BALANCER II
KADIE SOFTWARE
COPYRIGHT 1981 CARL KADIE

WHAT IS THE RAW EQUATION?
?_

9. Type in the equation, for example, H2 + O2 = H2O.
(directions for your own follow this summery)
Press [ENTER].

10. The screen will reprint your raw equation, then ChemBalancer II will get to work.
As it works, ChemBalancer II will flash various symbols in the upper right-hand corner of the screen. At first it flashs letter and numbers as it convert your raw equation to a mathematical representation. Next, it flashs a -< while it rearranges this new representation. Finally, it flashs and arrow (-→) while it does the actual balancing.

11. When ChemBalancer is finished, it will print the balanced equation. Using our example:

    2(H2) + (O2) → 2(H2O)

    Then it will ask:

    DO YOU HAVE ANOTHER EQUATION TO BALANCE?_

    Answer with YES or NO and press ENTER .

12. If you answer YES go back to step 8b on the previous page. If you answer NO, you may still balance other equations by following the instruction starting at step 8a on the previous page.

FORMAT NOTATION

Thoughout the examples given, this form will be used:

1. the raw equation as you would write it on paper. (raw means that there are no numbers in front of the chemicals)
2. the raw equation as you should type it to the computer.
3. the computer's response and approximate response time.

For example, an example may look like this:

1. $H_2 + O_2 \rightarrow H_2O$
2. H2 + O2 = H2O
3. 2(H2) + (O2) → 2(H2O) ½ min.

SIMPLE EQUATION

(ex #2) Preparation of Hydrogen

1. $Na + H_2O \longrightarrow NaOH + H_2$
2. NA + H2O = NA.O.H + H2
3. 2(NA) + 2(H2O) → 2(NA.O.H) + (H2) 1 min.

Notice that certain conventions are used because subscripts and lower-case letters are not available.

Looking at example 2, we see that each element symbol (e.g. Na, H, O) is made of either one or two letters. Any two letters will work, so you may use symbols that are not on the periodic table of the elements (see example 3).

Each chemical (e.g. Na, H_2O, NaOH, H_2) uses either a period (.) or a number to separate the symbols if there is more than than two (H2O, NA.O.H). Spaces may also be used; so that NA.O.H could be NA O.H or NA O H or, if you wish to use numbers, NA1O1H1. A final possibility is a combination of numbers, spaces, and periods (NA. O.H1, H2.O).

Subscripts are written after the element they refer to:

H_2 becomes H2

H_2O becomes H2O

If no number follows an element, the subscript is assumed to be one. So,

NA1.01.H1 is the same as

NA.O.H to ChemBalancer II.

The plus symbol (+) is used to separate chemicals on one side of the equation. The equal sign (=) is used to separate the two sides of the equation. The equal sign replaces the hand-written → , which means "yields" in the equation.

hand-written $H_2 + O_2 \rightarrow H_2O$

for the computer H2 + O2 = H2O

(notice also that blank spaces between chemical are not important.)

ADVANCED FEATURES

(ex #3) Hypothetical triple replacement

1. $A_3E_3I_2 + D_{21}H_8C_{21} + G_8B_{42}F_{21} \rightarrow A_2B_3C + D_2E_4F_3 + G_3H_2I_7$
2. A3E3I2 + D21.H8.C21 + G8.B42.F21 = A2.B3.C + D2.E4.F3 + G3.H2.I7
3. 28(A3E3I2) + 2(D21.H8.C21) + 3(G8.B42.F21) → 42(A2.B3.C) + 21(D2.E4.F3) + 8(G3.H2.I7)

5¾ min.

The advanced features included in ChemBalancer are for the most part internal. Multiple replacement balanceing, radical condensing, and mixed equation handling all fall into this category. Other advanced features however, are important externally, because they make life easier for you.

The first feature like this we shall look at, is element collecting. ChemBalancer II automatically collects like elements within a chemical, so you have more freedom. For example, H2O can be written as H.H.O, H.O.H, O.H.H, or O.H2; which ever you like best.

This feature is especially useful in organic chemistry. For example, Propene can be written :

C3H6, CC2H6, CCCH6, C2H4CH2, or many other ways, as long as there are three carbons and six hydrogens total in the chemical.

A second advanced feature is parentheses. You may nest parentheses up to 9 levels, so that glycerine can be written as either the awkward C3H503H3 or as the more conventional C3H5(O.H)3.

(ex #4) Preparation of soap from fat

1. $C_3H_5(C_{17}H_{35}COO)_3 + NaOH \rightarrow C_{17}H_{35}COONa + C_3H_5(OH)_3$
2. C3.H5.(C17.H35.C.O.O)3 + NA.O.H = C17.H35.C.O.O.NA + C3H5(O.H)3
3. (C3.H5.<C17.H35.C.O.O>3) + 3(NA.O.H) → 3(C17.H35.C.O.O.NA) + (C3H5 O.H 3)

2 1/3 min.

(ex #5) Hypothetical nested parentheses

1. $(((((((((A)_2)_4)_4)_2)_3)_6))_3) \rightarrow A$
2. (((((((((A)2)4)4)2)3)6))3) = A
3. (<<<<<<<<<(A>2>4>4>2>3>6>>3>) → 3456(A)

½ min.

Notice that in both examples the original parenthese were changed into <>, so that the program's own outer parentheses are distinct. Also notice that the subscript follows the right parenthesis just as it would follow an element.

OXIDATION-REDUCTION (Redox)

ChemBalancer II balances redox reactions by considering charges. Because superscripts are impossible on the TRS-80, and because the plus symbol is already used to signify the addition of chemicals, different notation must be used.

(ex #6) Cloride ion is oxidized to clorine gas

1. $MnO_4^- + H^+ + Cl^- \longrightarrow Mn^{2+} + Cl_2 + H_2O$
2. MNO4@- + H@ + CL@- = MN@2 + CL2 + H2O
3. 2(MNO4@-) + 16(H@) + 1Ø(CL@-) →
   2(MN@2) + 5(CL2) + 8(H2O) $3\frac{1}{2}$ min.

(note: MnO_4, could have, but doesn't need a period between the MN and the O4, because MN is two letters long.)

To summerize the charge notation:

CHARGE	WRITTEN NOTION	COMPUTER NOTATION
minus three	3- or ---	@-3
minus one	- or 1-	@- or @-1
positive one	+ or 1+	@ or @1
positive four	4+ or ++++	@4 or @@@@

Any negitive charge can be written as minus three is on the chart. Any positive charge can be written as positive four is on the chart. Remember the plus symbol (+) is never used in Balancer's notation.

(ex #7) Oxidation of iron

1. $Cr_2O_7^{2-} + H^+ + Fe^{2+} \rightarrow Cr^{3+} + Fe^{3+} + HOH$
2. CR2O7@-2 + H@ + FE@2 = CR@3 + FE@3 + H.O.H
3. (CR2O7@-2) + 14(H@) + 6(FE@2) →
   2(CR@3) + 6(FE@3) + 7(H.O.H)

3½ min.

In most cases it is best to put the charge at the end of the chemical as in examples 7 and 6, but the charge character (@) may be placed anywhere. So instead of CR2O7@-2 ,you may type (CR@6)2(O@-2)7

EXPLANATION OF ERROR MESSAGES

(ex #8) Example of error

1. ((((((((((A)))))))))) → A
2. ((((((((((A)))))))))) = A
3. FATAL ERROR: TOO MANY ('S 1/6 min.

Occurs when the number of left parentheses nested is greater than or equal 10.

(ex #9) Example of error

1. $H_2O \rightarrow H^+ + OH^-)_2$
2. H2O = H@ + O.H@-)2
3. FATAL ERROR: MISMATCHED ('S 1/6 min.

Occurs when the number of left parentheses is not equal to the number of right parentheses.

(ex #10) Example of error

1. $H_4 \rightarrow H_2 \rightarrow H$
2. H4 = H2 = H
3. TOO MANY ='S

1/12 min.

Occurs if there is more than one equal sign in the equation.

(ex #11) Example of error

1. ++ + → + +
2. ++ + = + +
3. FATAL ERROR: NO ELEMENTS GIVEN 1/30 min.

(ex #12) Example of error

1. $PbCl_2 \rightarrow AuCl_2$
2. PB.CL2 = AU.CL2
3. FATAL ERROR: ALCHEMY !!!!! 1/6 min.

Occurs when an element appears on only one side of the equation. When testing for this error, ChemBalancer II considers negative charges as if they were on the other side of the equal (yield) sign.

(ex #13) Example of error

3. FATAL ERROR: CAN'T BE BALANCED

No known examples

(ex #14) Example of error

1. $AB + C_2D \rightarrow AD + CB$
2. A.B + C2D = A.D + C.B
3. FATAL ERROR:
   MULTIPLE REPLACEMENT REACTION, CAN'T BE BALANCED 1 min.

Many replacement reactions, like the one in this example, are impossible to balance; however, others only very difficult. ChemBalancer II can tell the difference and will balance all the others. When this message is printed, it means that the equation can never be balanced without violating the law of conservation of matter (and sometimes charge).

(ex.) A + B.C = A.B.C2

This type of error, also causes the MULTIPLE REPLACEMENT REACTION, CAN"T BE BALANCED message to be printed.

(ex #15) Example of error

1. $H_2 \longrightarrow H!$
2. H2 = H!
3. FATAL ERROR: ILLEGAL CHARACTER 1/60 min.

Some character, besides the letters A-Z, the numerals Ø-9, and the symbols +,=,. and @ appear in the equation.

(ex #16) Example of error

1. (((((($H_{1ØØØ})_{1ØØØ})_{1ØØØ})_{1ØØØ})_{1ØØØ})_{1ØØØ})_{1ØØØ} \longrightarrow H$
2. ((((((H1ØØØ)1ØØØ)1ØØØ)1ØØØ)1ØØØ)1ØØØ)1ØØØ = H
3. OV ERROR IN 41Ø $\frac{1}{4}$ min.

or 3. FATAL ERROR: OVERFLOW

The numbers of the reaction exceed the limit of the computer or ChemBalancer II.

(ex #17) Example of error

1. $H_2SO_4 + K \longrightarrow HK + KSO_4 + H_2$
2. H2.S.O4 + K = H.K + K.S.O4 + H2
3. FATAL ERROR: AMBIGUOUS EQUATION 1½min.

This means there is more than one possible solution and/or that one or more of the chemicals is unnecessary. In this example either H_2 or HK could be left out and it could still be balanced, or it could be balanced:

$2\ H_2SO_4 + 4K \longrightarrow 2HK + 2KSO_4 + H_2$ or

$3\ H_2SO_4 + 5K \longrightarrow 2HK + 3KSO_4 + 2H_2$

This is perhaps the most important message, because when balancing by hand with trial and error, there is a great danger that the ambiguity will not be detected.

(ex #18) Example of error

1. $H_2 \longrightarrow 2H$
2. H2 = 2H
3. FATAL ERROR: SUBSCRIPT IN INCORRECT POSITION 1/20 min.

The only numbers allowed in the raw equation are subscripts, and they must follow elements or a right parenthesis.

(ex # 19) Example of error

1. ABC + CB + $C_2B \longrightarrow AC_{17}B$
2. A.B.C + C.B + C2B = A.C17B
3. FATAL ERROR: NEGATIVE SOLUTION

1 1/6 min.

The only way to balance this type of equation would be to supply negitive answers. Like:

ABC + -16 CB + 16 $C_2B \longrightarrow AC_{17}B$

(ex #20) Formation of phosphorus cloride and decomposition of water

1. P + H_2O + $Cl_2 \longrightarrow H_2$ + PCl_5 + O_2
2. P + H2O + CL2 = H2 + P.CL5 + O2
3. WARNING MIXED EQUATIONS

   2(P) + 2(H2O) + 5(CL2) → 2(H2) + 2(P.CL5) + (O2)

1 min.

In a mixed equation two or more non-related equation are given to ChemBalancer II. As soon as it detects this condation it prints the warning, then it balances both (or all if there is three or more) and prints out the answer.

* * * *

This ends the instructions. If you have any orders, suggustions, questions, or problems please write to:

Kadie Software

[redacted]

Peoria, IL 61603

Unbalanced equations

Digestion of calcium phospate with dilute sulfric acid.

$$Ca_3(PO_4)_2 + H_2SO_4 + H_2O \rightarrow H_3PO_4 + CaSO_4(H_2O)_2$$

Preparation of chloroform.

$$CCl_4 + Fe + H_2O \rightarrow CHCl_3 + Fe_3O_4 + FeCl_2$$

Oxidation of sugar.

$$KCLO_3 + C_{12}H_{22}O_{11} \rightarrow KCL + CO_2 + H_2O$$

Careful heating of sodium sulfite.

$$Na_2SO_3 \rightarrow Na_2SO_4 + NaS$$

Combustion of coal or wood.

$$C_6H_{1\emptyset}O_5 + O_2 \rightarrow CO_2 + H_2O$$

Combustion of gasoline.

$$C_8H_{18} + O_2 \rightarrow CO_2 + H_2O$$